

Science Olympiad
Chemistry Lab C Test (**CLASS SET**)
2025 ATX Science Olympiad Invitational



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Directions:

- Each participant may bring 1 sheet of notes, a standalone class III calculator, and lab equipment as specified in the rules manual.
- Please pay attention to all lab instructions given out during the event. Failure to comply and not cleaning up afterwards will result in a penalty up to 10% of the exam score.
- This was made using LastLegume's Scioly Test Template. Thanks for the help!
- You will have 50 minutes to complete this exam - do not begin until instructed to do so!
- All answers must be indicated on your answer document. **DO NOT WRITE ON THIS EXAM!**
- Point values are indicated for all questions & tiebreakers are indicated in test booklets. The number of tiebreakers correctly answered indicates your tie order.
- You may take apart the test to split the work, but make sure to staple it back together at the end of the event.
- Good luck on the test!

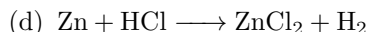
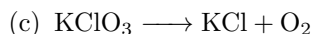
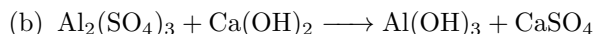
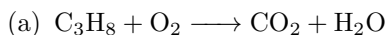
Reaction Identification Multiple Choice (7 points)

1. Classify the following chemical reactions. Circle the best answer for each.

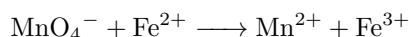
- (a) (1 point) $\text{NH}_4\text{NO}_3(\text{s}) \longrightarrow \text{N}_2\text{O}(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
- A. Decomposition
 - B. Disproportionation
 - C. Combustion
 - D. Double displacement
- (b) (1 point) $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \longrightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$
- A. Single displacement
 - B. Decomposition
 - C. Redox
 - D. Double displacement
- (c) (1 point) $\text{Cu}(\text{s}) + 2\text{AgNO}_3(\text{aq}) \longrightarrow \text{Cu}(\text{NO}_3)_2(\text{aq}) + 2\text{Ag}(\text{s})$
- A. Redox
 - B. Comproportionation
 - C. Hydrolysis
 - D. Double displacement
- (d) (1 point) $\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{l})$
- A. Combustion
 - B. Single Displacement
 - C. Decomposition
 - D. Neutralization
- (e) (1 point) $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \longrightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- A. Double displacement
 - B. Synthesis
 - C. Acid–base
 - D. Redox
- (f) (1 point) $\text{C}_{10}\text{H}_{22}(\text{l}) + \text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$
- A. Synthesis
 - B. Decomposition
 - C. Double displacement
 - D. Redox
- (g) (1 point) $\text{H}_2\text{S}(\text{g}) + \text{SO}_2(\text{g}) \longrightarrow \text{S}(\text{s}) + \text{H}_2\text{O}(\text{g})$
- A. Decomposition
 - B. Double displacement
 - C. Acid–base
 - D. Comproportionation

Reaction Balancing (19 points)

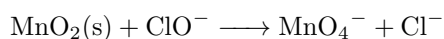
2. (4 points) Balance the following chemical reactions using lowest whole-number coefficients. Write the coefficients in order.



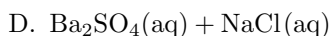
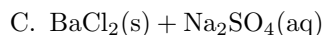
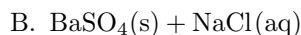
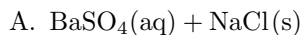
3. (3 points) Balance the following redox reaction in acidic conditions. Provide the balanced equation with lowest whole-number coefficients.



4. (4 points) Balance the following redox reaction in basic conditions. Provide the balanced equation with lowest whole-number coefficients.



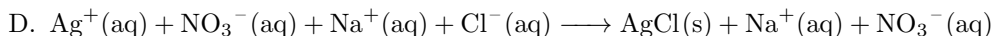
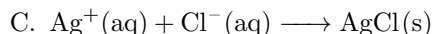
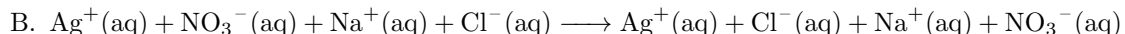
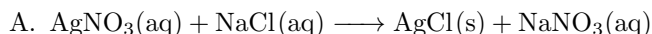
5. (2 points) Barium chloride and sodium sulfate salts are added to a beaker of water. After some time, a white solid appears. What products are associated with this reaction?



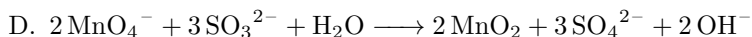
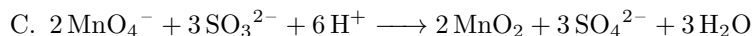
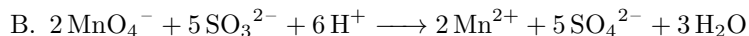
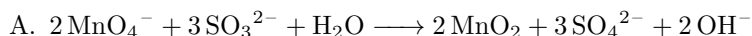
6. (2 points) Which salt could be added to water to effectively remove phosphate?



7. (2 points) Which is the correct total ionic equation for the reaction between aqueous silver nitrate and sodium chloride?



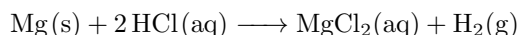
8. (2 points) When acidified potassium permanganate solution is added to sodium sulfite solution, what is the balanced net ionic reaction?



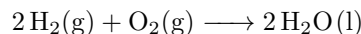
Stoichiometry (14 points)

9. (2 points) A 0.25 L solution of 0.15 M NaCl is mixed with 0.5 L of 0.55 M MgCl₂. What is the concentration of chloride in the final solution?
- A. 0.78 M
B. 0.41 M
C. 0.69 M
D. 0.83 M
10. (2 points) How many chlorine atoms are in 400 grams of CH₂Cl₂?
- A. 4.71×10^{24}
B. 5.67×10^{24}
C. 2.84×10^{24}
D. 1.42×10^{25}

11. (2 points) 12.0 g of magnesium is reacted with 550 mL of 1 M hydrochloric acid to produce hydrogen gas:



The hydrogen gas is then collected and reacted with oxygen to produce water:



If 4.0 g of water is actually obtained, what is the percent yield of water?

- A. 55%
B. 45%
C. 81%
D. 41%
12. (8 points) A multi-part problem on an unknown organic compound X (C, H, O only).
You are given an unknown organic compound X that contains only C, H, and O. The following experiments are done to figure out its formula.
- Combustion experiment: A 0.5000 g anhydrous sample of X is burned completely. The products are 0.880 g CO₂ and 0.180 g H₂O.
 - Mass spectrometry: The molar mass of anhydrous X is determined to be 100 g mol⁻¹.
 - Hydrate Test: A 2.000 g sample of crystalline X is heated to remove water of hydration. After heating, only 1.4706 g of the dry compound remains. (Assume only water is lost.)

Answer the following:

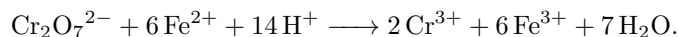
- (a) Determine the moles of C, H, and O in a 0.5000 g sample of X.
(b) Determine the empirical formula of X.
(c) Using the molar mass, what is the molecular formula of X?
(d) From the hydrate data, determine the value of *n* in X · *n* H₂O.

Kinetics Multiple Choice (18 points)

13. (3 points) Which statement best aligns with the principles of collision theory?
- A. Catalysts increase reaction rates by increasing the frequency of collisions between reactant molecules.
 - B. For a reaction to occur, reactant molecules must collide with proper orientation.
 - C. Increasing temperature decreases the frequency of collisions by causing molecules to move past each other more quickly.
 - D. Grinding a solid reactant into a powder slows the reaction because larger solid blocks provide more reactive surface area.
14. (3 points) The concentration of reactants is recorded over time for the irreversible reaction $A \longrightarrow B$. It is observed that the half-life is decreasing as the reaction progresses. Which of the following could describe the rate law of the reaction?

I. $\text{Rate} = k[A]$ II. $\text{Rate} = k[A]^2$

- A. I only
 - B. II only
 - C. Both I and II
 - D. Neither I nor II
15. (3 points) Iron(II) ions are oxidized by acidified potassium dichromate according to the balanced reaction



Which of the following best describes the relationship between $\frac{d[\text{Fe}^{2+}]}{dt}$ and $\frac{d[\text{Cr}^{3+}]}{dt}$?

- A. $6\frac{d[\text{Fe}^{2+}]}{dt} = 2\frac{d[\text{Cr}^{3+}]}{dt}$
- B. $6\frac{d[\text{Fe}^{2+}]}{dt} = -2\frac{d[\text{Cr}^{3+}]}{dt}$
- C. $2\frac{d[\text{Fe}^{2+}]}{dt} = 6\frac{d[\text{Cr}^{3+}]}{dt}$
- D. $2\frac{d[\text{Fe}^{2+}]}{dt} = -6\frac{d[\text{Cr}^{3+}]}{dt}$

16. (3 points) A given reaction has reactants A, B, and C. The following experimental data were obtained from multiple trials at the same temperature. What is the value of the rate constant k for this reaction?

Experiment	[A] (M)	[B] (M)	[C] (M)	Initial rate (M s ⁻¹)
1	0.100	0.050	0.100	1.00×10^{-4}
2	0.100	0.100	0.100	2.00×10^{-4}
3	0.100	0.200	0.200	8.00×10^{-4}
4	0.300	0.050	0.100	1.00×10^{-4}

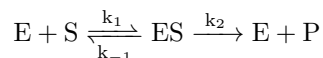
- A. $2.0 \times 10^{-2} \text{ M}^{-1}\text{s}^{-1}$
B. $4.0 \times 10^{-1} \text{ M}^{-1}\text{s}^{-1}$
C. $1.0 \times 10^{-3} \text{ M}^{-1}\text{s}^{-1}$
D. $1.0 \times 10^{-2} \text{ M}^{-1}\text{s}^{-1}$
17. (3 points) The rate constant k for a certain reaction is $1.50 \times 10^{-5} \text{ s}^{-1}$ at 200°C and $2.50 \times 10^{-3} \text{ s}^{-1}$ at 400°C . What is the activation energy E_a for this reaction? (Assume E_a and A are constant.)
A. 7.39 kJ mol^{-1}
B. 17.0 kJ mol^{-1}
C. 29.4 kJ mol^{-1}
D. 67.7 kJ mol^{-1}
18. (3 points) The reversible exothermic reaction $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$. When the temperature increases, less product is present at equilibrium. Which of the following best describes how the rate constants of the forward and reverse reactions, k_1 and k_{-1} , change as temperature increases?
A. k_1 increases, k_{-1} decreases
B. k_1 decreases, k_{-1} increases
C. k_1 increases, k_{-1} increases
D. k_1 decreases, k_{-1} decreases

Kinetics Free Response (40 points)

An Introduction to Enzyme Kinetics

Traditionally, chemistry is taught with a strong emphasis on thermodynamics and equilibrium. While thermodynamics predicts what will happen in the long run, kinetics explains what actually happens in the moment. For slow reactions, kinetics are most important, and the discussion of equilibrium becomes largely irrelevant. As the economist John Maynard Keynes famously said, “In the long run, we are all dead.”

Catalysts play a particularly crucial role in biological systems. In biology, these catalysts are known as enzymes. The study of enzyme kinetics, particularly Michaelis–Menten kinetics, is fundamental to understanding biochemistry. Typically, the conversion of a substrate (reactant) into a product by an enzyme follows the mechanism with two elementary steps described below:



19. (5 points) On your answer sheet, you will find a graph of energy vs. reaction progress for the uncatalyzed reaction $\text{S} \longrightarrow \text{P}$. On the same diagram, draw a curve describing the energy vs. reaction progress for the catalyzed reaction that proceeds through the mechanism above.

20. Derivation of the Michaelis–Menten equation

- (2 points) First, we will assume that the concentration of the intermediate complex, $[\text{ES}]$, is constant. What is the name of this assumption?
- Let k_1 and k_{-1} denote the forward and reverse rate constants for the reversible first step, respectively. Let k_2 denote the rate constant for the irreversible second step.
 - (2 points) Write an equation for the rate of formation of $[\text{ES}]$.
 - (2 points) Write an equation for the rate of decomposition of $[\text{ES}]$.
 - (2 points) Using the assumption that $[\text{ES}]$ is constant, relate these two equations with a single equation.
 - (2 points) Define $[\text{E}]_t$ as the total enzyme concentration. Express $[\text{E}]$ (free enzyme) in terms of $[\text{E}]_t$ and $[\text{ES}]$.
 - (4 points) Substitute your expression from part (iv) into part (iii). Now solve for $[\text{ES}]$.
 - (3 points) Write an equation for the rate of formation of product $V = \frac{d[\text{P}]}{dt}$ in terms of k_2 and $[\text{ES}]$. Then substitute your expression for $[\text{ES}]$.
 - (3 points) Define $V_{\max} = k_2[\text{E}]_t$ and $K = \frac{k_{-1} + k_2}{k_1}$. Rewrite the rate law in terms of $V_{\max}$, K , and $[\text{S}]$.

21. Rate law interpretation and graphical analysis

- (a) (4 points) Based on both the graph and your derived rate law, fill in the blanks (on your answer sheet!) with “0th”, “1st”, or “2nd”. If necessary, you may use any answer choice more than once.

The reaction is _____ order with respect to $[\text{S}]$ at low concentrations, and _____ order with respect to $[\text{S}]$ at high concentrations.

- (3 points) Describe at a molecular level what is happening at low substrate concentrations. (Two sentences or fewer.)
- (3 points) Describe at a molecular level what is happening at high substrate concentrations. How does this relate to $V_{\max}$? (Three sentences or fewer.)
- (5 points) Imagine another substrate, S' , is present in solution at a moderate concentration and competes with S for binding to E . On the same diagram of V vs. $[\text{S}]$, sketch qualitatively what the curve would look like with S' present, and label it clearly.

Laboratory Practical (60 Points)

Today, you will be performing a variant of the famous iodine clock reaction. This experiment runs with two reactions happening simultaneously:

- Hydrogen peroxide (H_2O_2) oxidizes I^- to I_2 in acidic solution.
- Thiosulfate ($\text{S}_2\text{O}_3^{2-}$) immediately reduces I_2 back to I^- .

At first, both of these reactions compete, with the thiosulfate reaction preventing visible I_2 accumulation. However, once the thiosulfate runs out, I_2 begins to build up, forming a complex with starch and producing a sharp color change.

Your task: determine the reaction order with respect to $[\text{I}^-]$.

Materials:

- 100 mL of aqueous 0.040 M KI
 - 100 mL of aqueous starch + 0.020 M $\text{Na}_2\text{S}_2\text{O}_3$
 - 100 mL of aqueous 0.20 M H_2SO_4
 - 100 mL of aqueous 0.10 M H_2O_2
 - 3 $\times$ 50 mL beaker
 - 2 $\times$ 10 mL graduated cylinder
 - Plastic pipettes
 - Deionized water
 - Stopwatch
- (4 points) Write balanced net ionic equations for each of the two reactions given above.
 - (6 points) What is the relationship between the time it takes for the color to change and the rate of iodine production? Why?
 - (10 points) Provide a brief description of your experimental procedure. Note that you must have at least five data points for full credit.
 - (10 points) Record your data and observations.
 - (20 points) Show all calculations. You do not need to graph your data in order to receive full credit, though doing so may be useful.
 - (10 points) What is the reaction order with respect to $[\text{I}^-]$?